

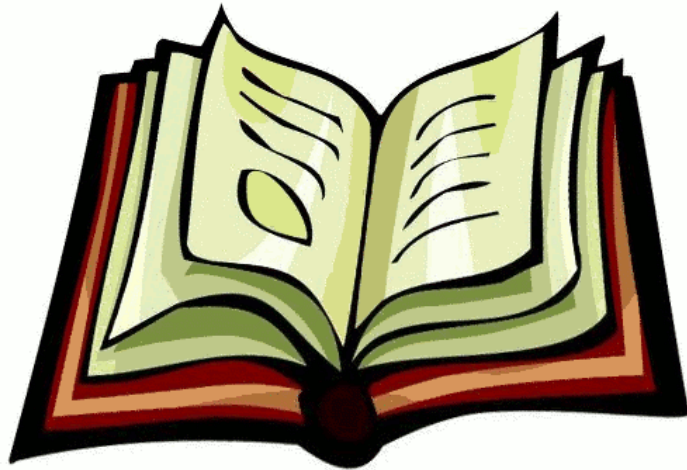


Exploring Data

F28ED

The Challenge

Find a **meaningful** and **accurate** way to **depict** the '**true story**' of data and clearly **report** the data's **main features**

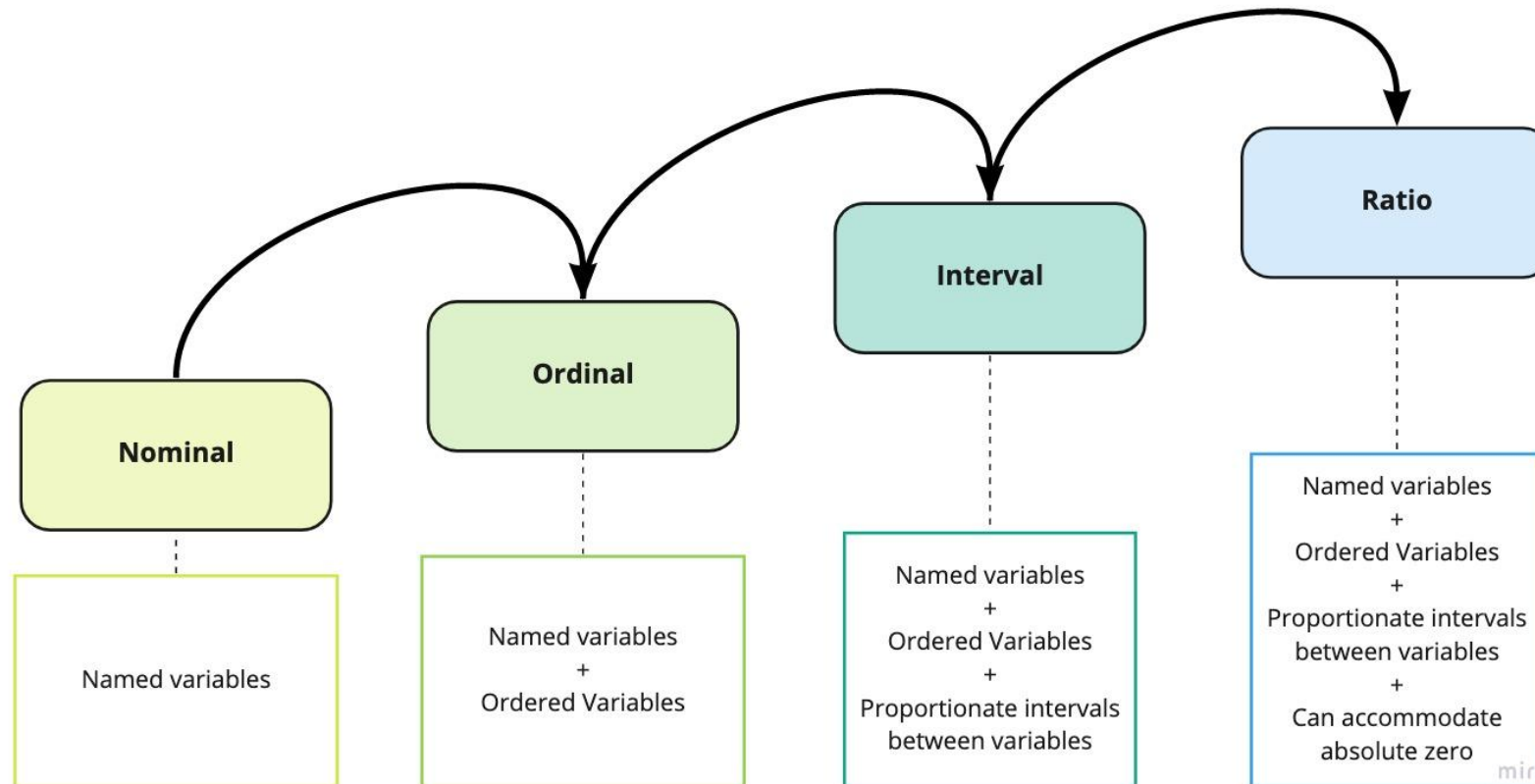




Get your fingers dirty
with data

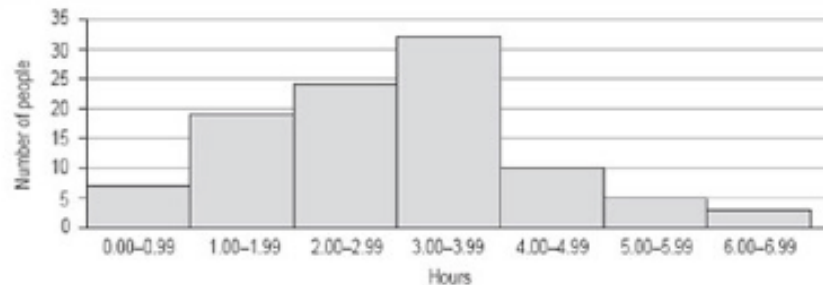
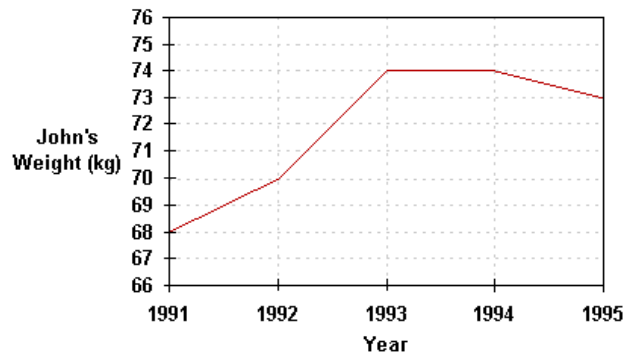
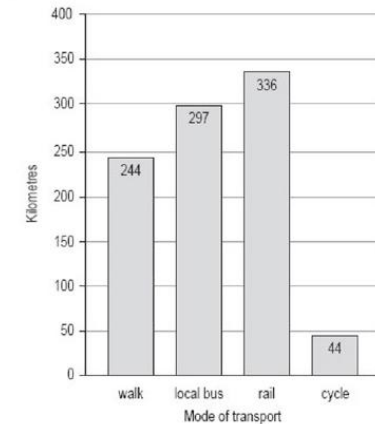
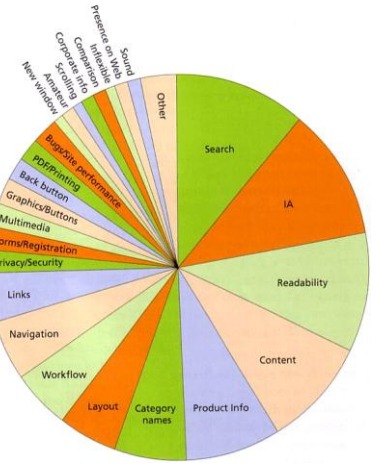
In the previous lecture...

- Measurement is important



In the previous lecture...

- We can show **discrete** and **continuous** variable data in **tables**
- We can show **discrete** data in **bar charts** and **pie charts**
- We can show **continuous** variable data in **line graphs** and **histograms**

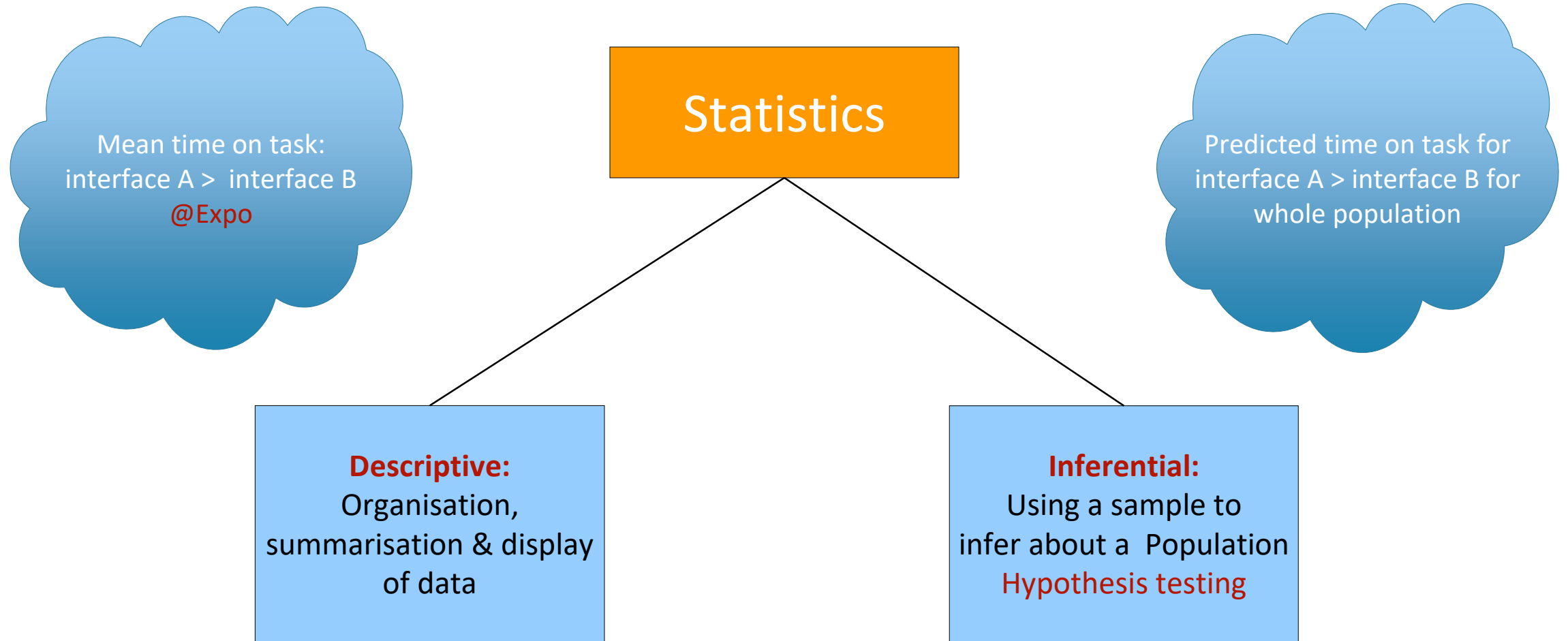


In this lecture

- Summarising or 'describing' data
- Applying what you've learnt to your CW Report



Descriptive vs Inferential Statistics



Descriptive Statistics

- Why use descriptive statistics?
- Measures of Central Tendency
 - ▶ Mode
 - ▶ Median
 - ▶ Mean
- Measures of Spread
 - ▶ Range
 - ▶ Standard Deviation
 - ▶ Variance

What do we want to describe?

- Descriptive statistics:
 - summarise a set of data into just a couple of numbers that represent the entire data set

Descriptive Statistics: Example

Since someone told us that the **mean shoe size** for a man is 9, we borrow money and buy 1000 pairs of size 9 shoes

What happens?

You go broke! :(



Descriptive Statistics: Example

You try again, but this time you borrow money and buy 1100 pairs of shoes, 100 of each of size 4 - 14

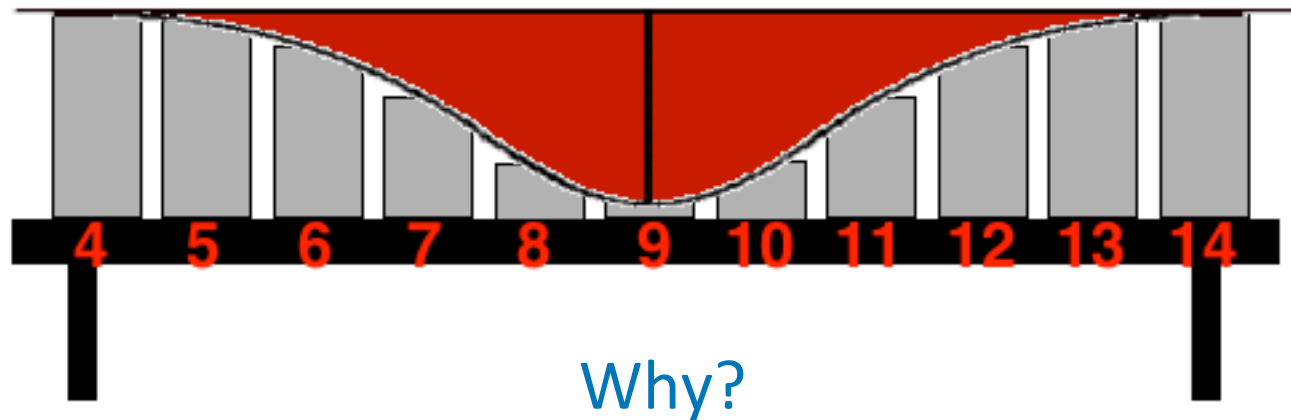
What happens?

You still go broke! :(



Descriptive Statistics: Example

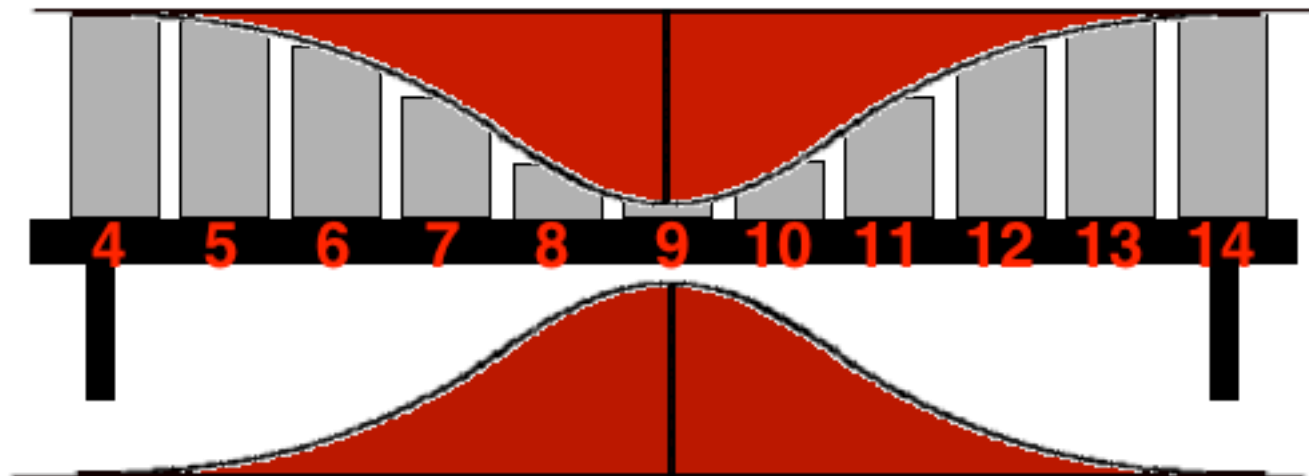
After a few weeks of being open, here is what your stock of shoes looks like:



You did not take into account that there are
not equal numbers of feet of each size

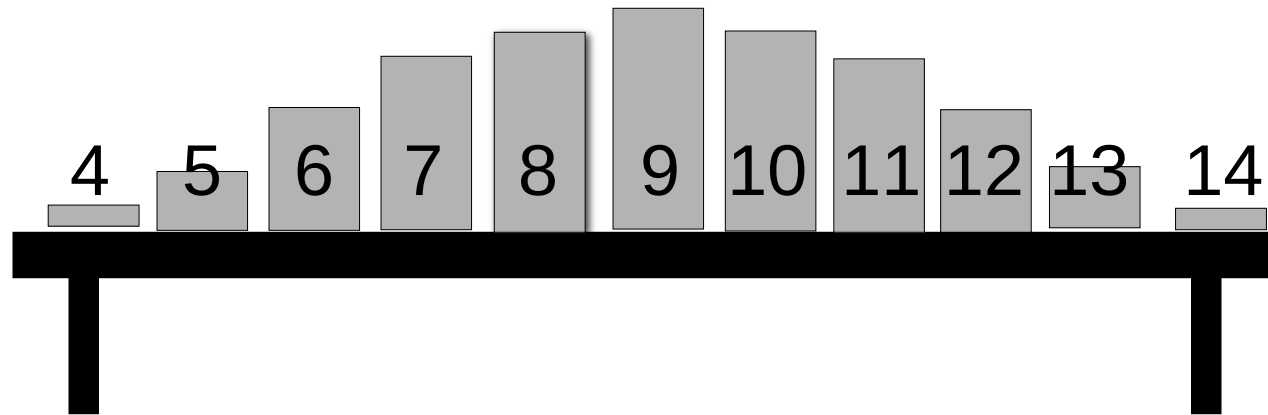
Descriptive Statistics: Example

After a few weeks of being open, here is what your stock of shoes looks like:



Notice that this is an **inverted normal** distribution and the **mean** is at size 9

Descriptive Statistics: Example



Conclusion: to buy the correct amount of stock, you need to know the typical score – the **central tendency** and how the scores are spread out – the **variability**

Measures of central tendency

- Single number that is used to represent the general magnitude of scores in the data set
- Representative value of the set
 - Mode
 - Median
 - Mean

Mode

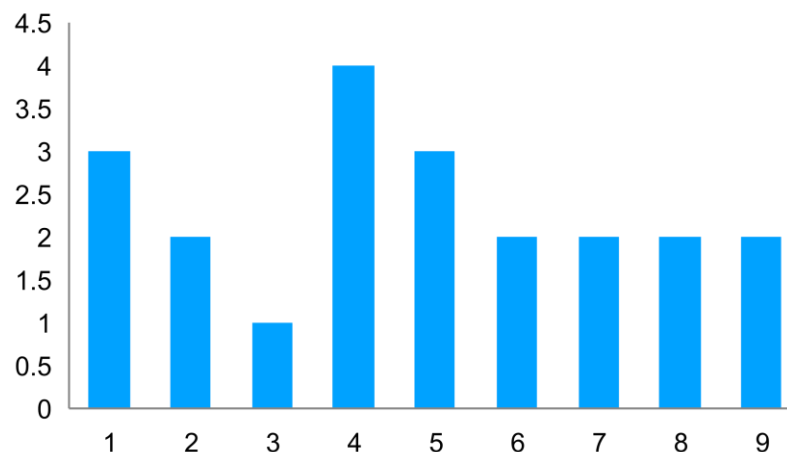
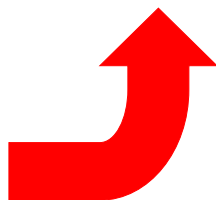
- The most frequent score

Example: 5 9 7 4 6 8 2 4 1 3 5 1 4 6 9 8 7 5 2 4 1

Ordered: 1 1 1 2 2 3 4 4 4 4 5 5 5 6 6 7 7 8 8 9 9

Frequency count: 3 2 1 4 3 2 2 2 2

Mode



Mode

- There is **no mode** when all the **scores** are **different** (or there is the same number of many scores)
 - E.g., 2 5 7 9 11 1 8 0
- Sometimes there is **more than one** mode
 - E.g., 2 5 7 **9 9** 11 **1 1** 8 0
- **Limitation**
 - Does **not take into account other scores** so comments about distribution may be misleading

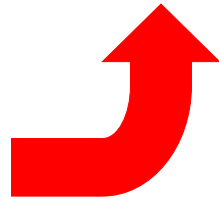
Median

- The middle score (of an ordered set)

Example (odd # of scores):

3 8 11 11 12 13 24 35 46

Median



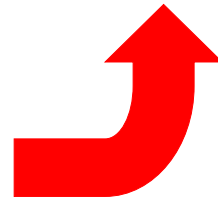
Median

- The middle score (of an ordered set)

Example (even # of scores):

3 8 11 11 12 13 24 35 46 48

Median



$$\frac{12 + 13}{2} = 12.5 = \textit{median}$$

- Limitation
 - Does not take into account extreme scores

Mean

- The arithmetic **average** of scores
 - **mathematical centre** of the **distribution** of scores
 - **BUT** it is **not the middle** score
 - it is the **computed centre**

$$\text{Mean: } \bar{x} = \frac{1}{N} \sum_i x_i$$

Mean

- The arithmetic **average** of scores

$$\bar{x} = \frac{1}{N} \sum_i x_i$$

Example: 3 8 11 11 12 13 24 35 46 48

$$\bar{x} = \frac{3 + 8 + 11 + 11 + 12 + 13 + 24 + 35 + 46 + 48}{10}$$

$$\bar{x} = \frac{211}{10} = 21.1$$


Mean

Beware of inappropriate averaging...

With your head in an oven
and your feet in ice
you would feel,
on average,
just fine



The majority of people have more than the average number of legs
(**Mean** = 1.9999)



Levels of Measurement

Nominal - data that is **classified** into **categories** and **cannot be arranged** in any particular **order**.

EXAMPLES: eye colour, gender, pass/fail

Ordinal – involves data **arranged** in some **order**, but the **differences** between data values **cannot be determined** or are meaningless.

EXAMPLES: Language ability: Beginner; Intermediate; Fluent

Level of agreement: Strongly disagree; Disagree; Neither agree nor disagree; Agree; Strongly agree

Interval - similar to ordinal, with the additional property that meaningful amounts of **differences** between data values can be **determined**. There is **no natural zero point**.

EXAMPLES: Temperature on the Celsius scale

Ratio - interval with an inherent **zero starting point**. **Differences** and **ratios** are **meaningful** for this level of measurement.

EXAMPLES: Kelvin, height, weight, time

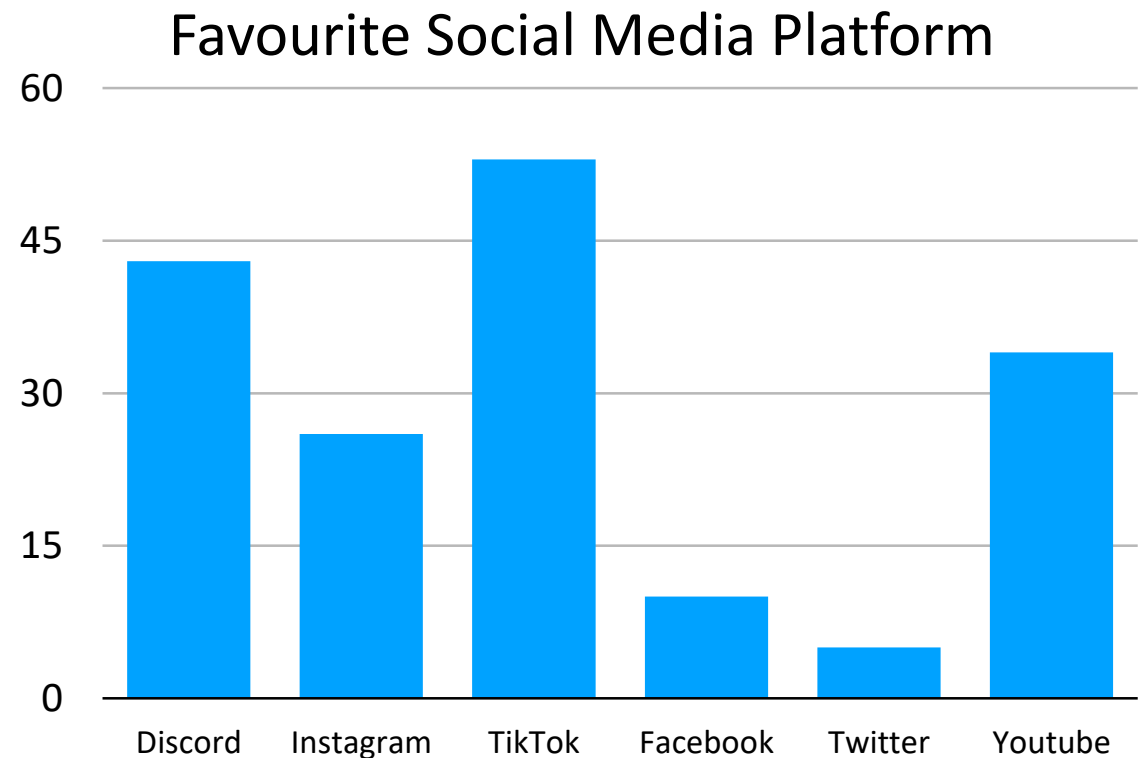
What statistics to use?

Data type	Reporting Central Tendency
Nominal	Mode
Ordinal	Mode or Median
Interval and ratio	Mode, Mean or Median

What statistics to use? Nominal

- If **Nominal**: you **must** use the **mode**
- Mode represents the **most popular**

	Favourite social media platform
Discord	43
Instagram	26
TikTok	53
Facebook	10
Twitter	5
Youtube	34
Total	128



What statistics to use? Ordinal

- If **Ordinal**: you **can** use either the **mode** or the **median**
- **No averaging** of **Likert scale** responses!



1



2



3



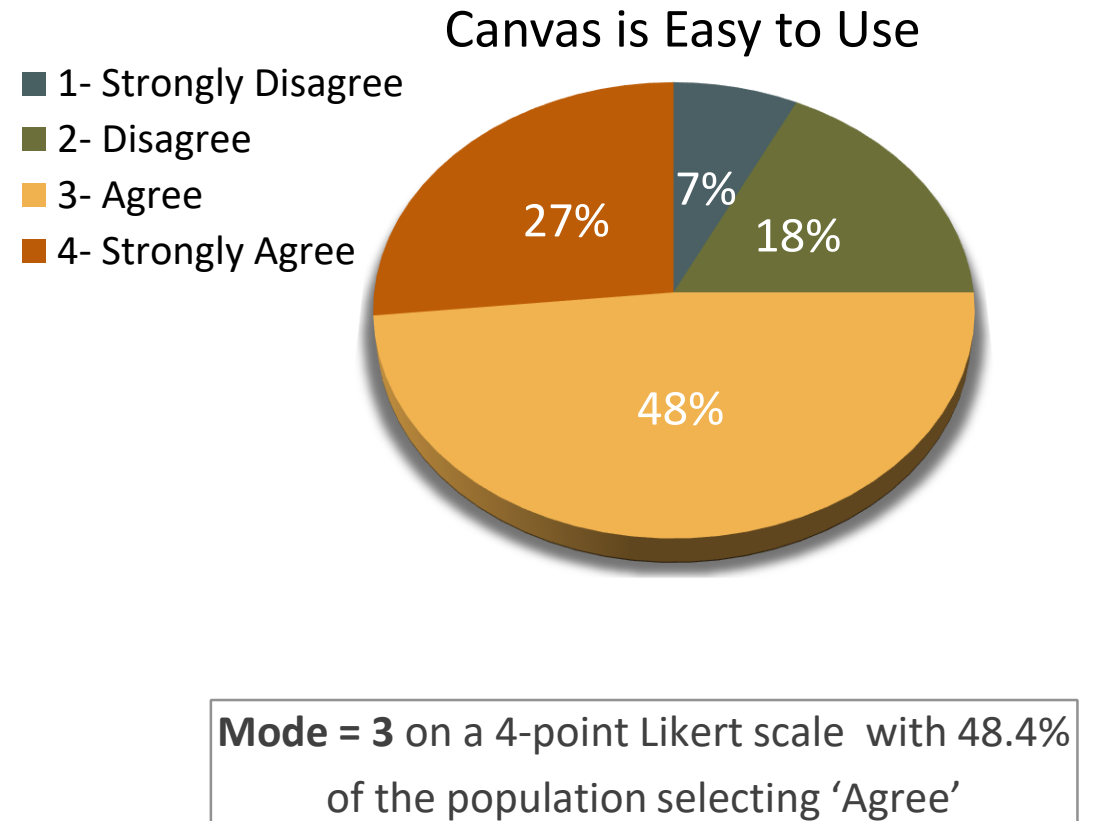
4



5

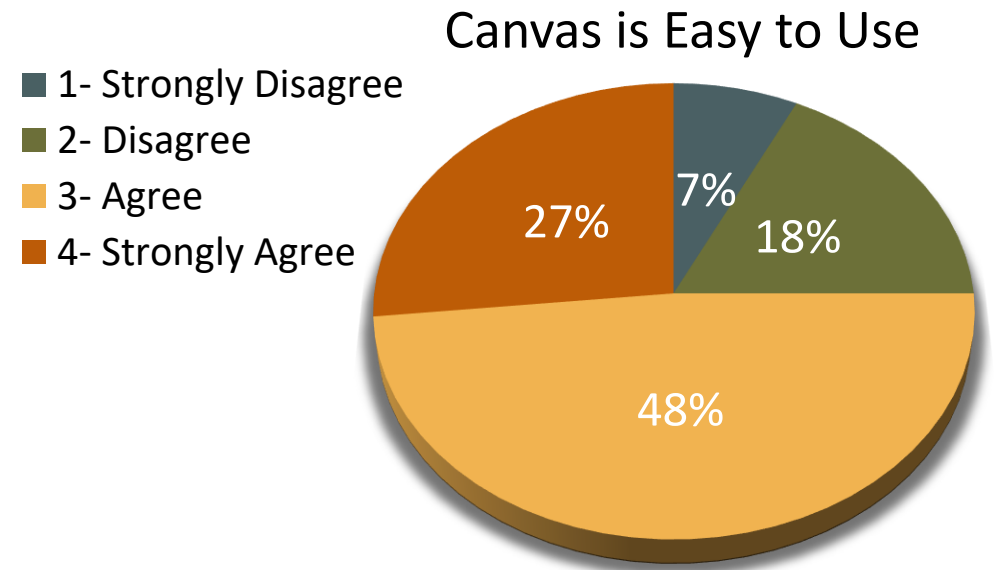
What statistics to use? Ordinal

- **Mode** = the **most popular** response
- Examples
 - Mode = 4, i.e., 80% of participants rated the game as 4 on a 5-point rating scale
 - 60% of females compared to 80% of males rated the interface 6 or above on a 7-point rating scale



What statistics to use? Ordinal

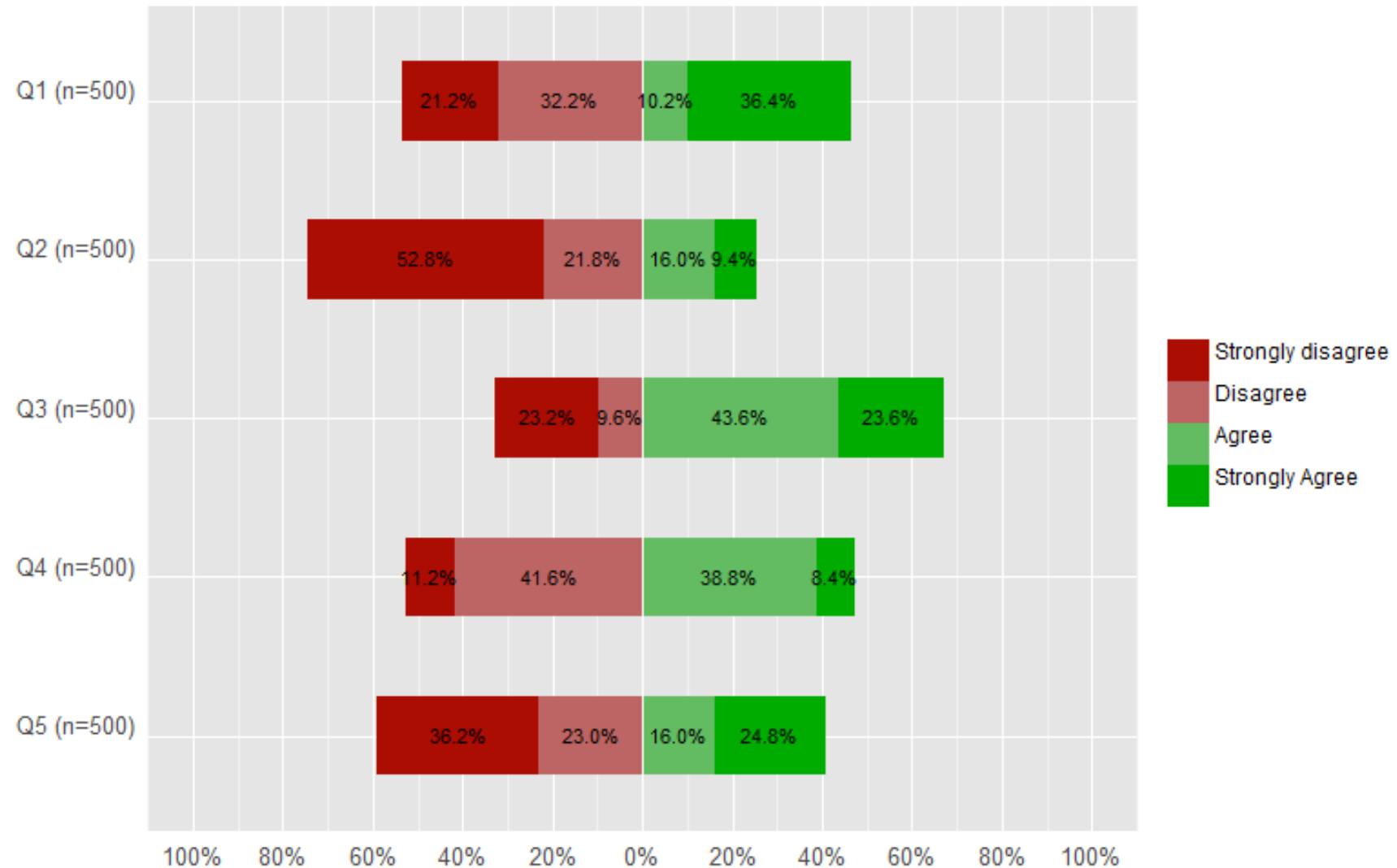
- Median



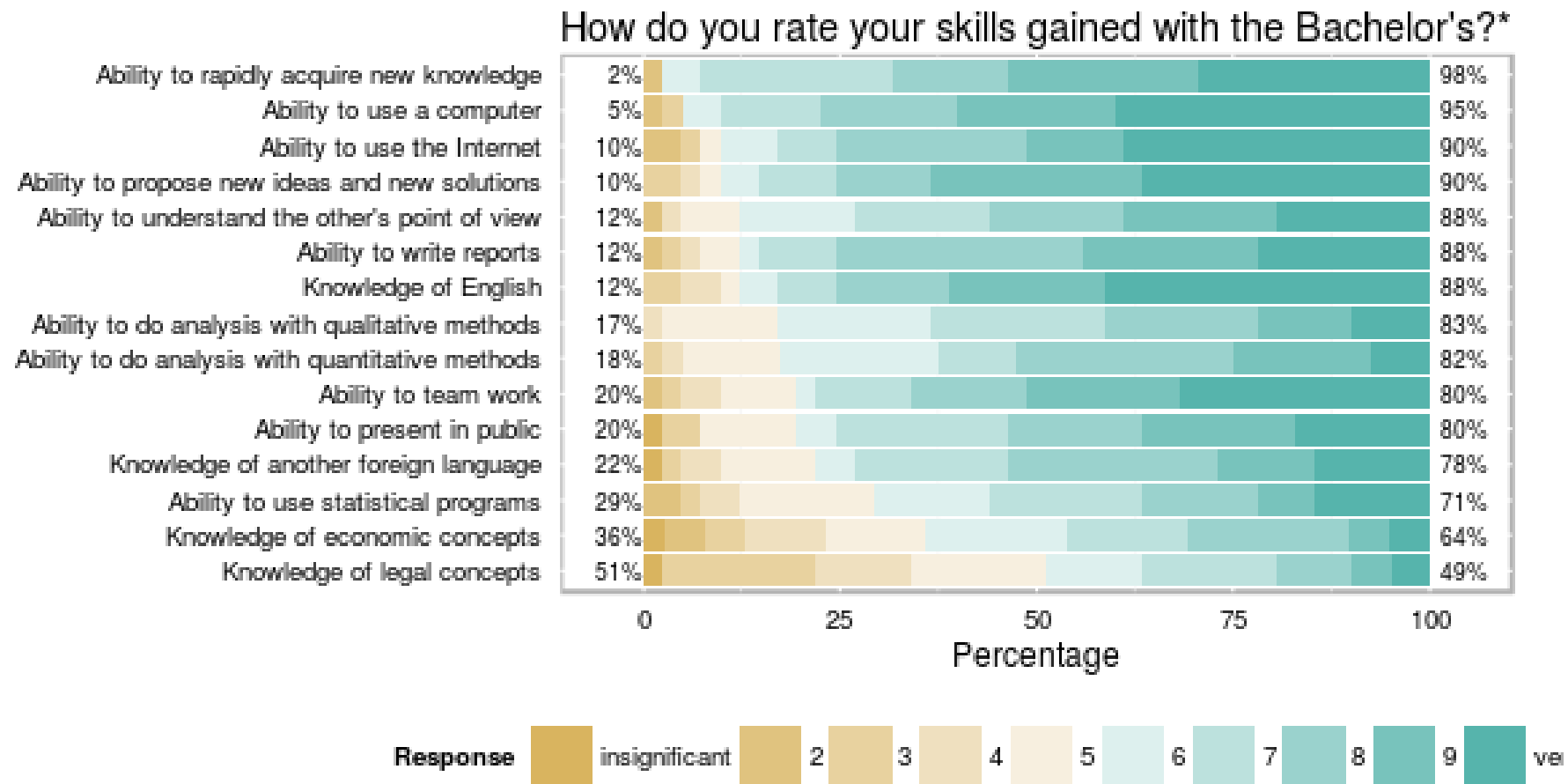
11111111222222222222222222223333333333333333333333333333
3333333333333333333333333333333344444444444444444444444444

Median = 3

Likert scales visualisation



Ordinal data: Rating scales



What Statistics to Use? Interval / Ratio

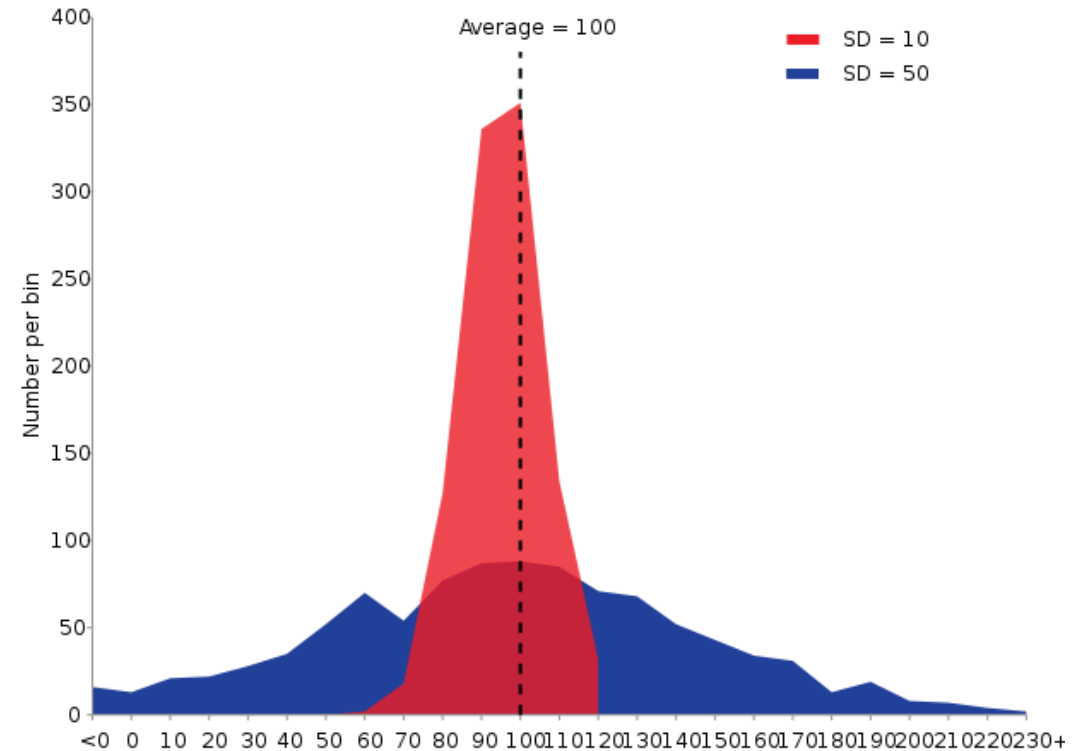
- If **Interval** or **Ratio**: use any but the **mean** is the most typical
- Examples
 - The average time to complete the task is 15.5 seconds
 - The average temperature in Edinburgh in November is 7 degrees Celsius :(

Descriptive Statistics

- Why use descriptive statistics?
- Measures of Central Tendency
 - ▶ Mode
 - ▶ Median
 - ▶ Mean
- Measures of Spread
 - ▶ Range
 - ▶ Standard Deviation
 - ▶ Variance

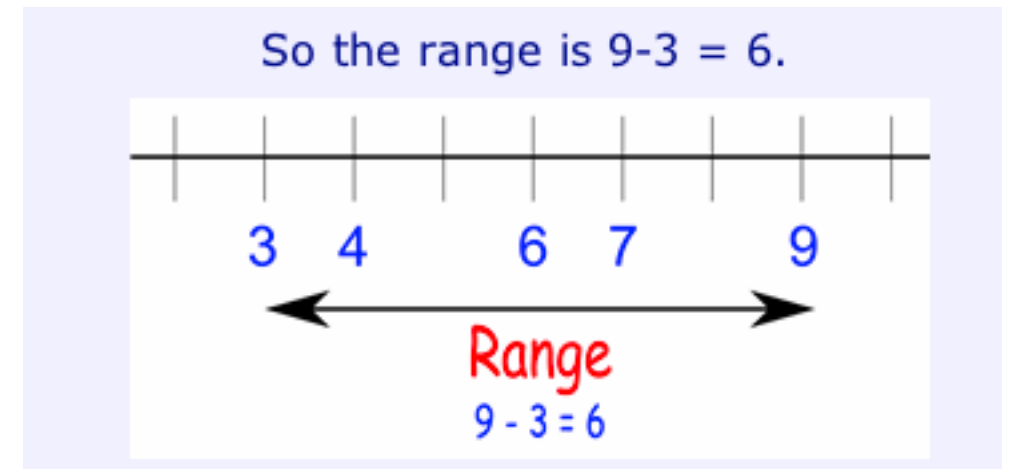
Dispersion: Range, variance and Standard deviation

- Average time on task:
blue = men and red = women
- They have equal mean and different standard deviations
 - What does this tell you?



Dispersion: Range

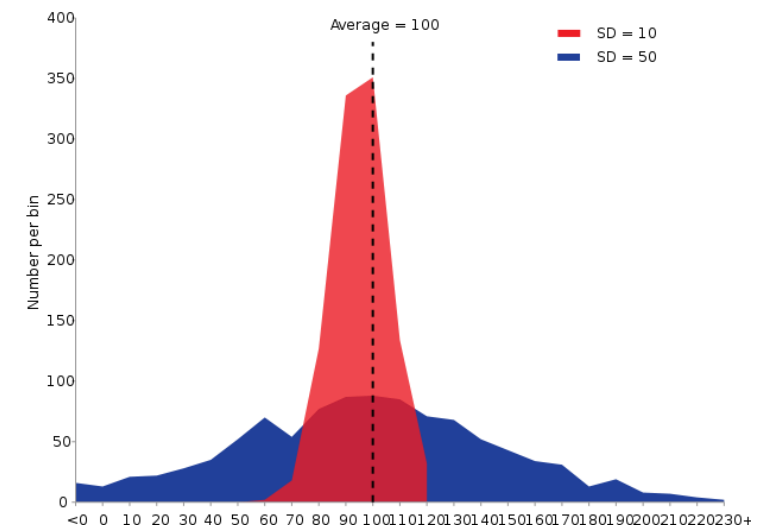
- **Range**: **difference** between the **lowest** and the **highest** number
- Example: {4,6,9,3,7}
- Range can be **misleading**. E.g., {8, 11, 5, 9, 7, 6, 3616}
 - **Lowest** value: 5; **highest** value: 3616
 - Range is $3615 - 5 = 3611$



Variance/Standard Deviation

- Variance captures the spread of numbers:
 - How far the numbers are from the mean
- Standard deviation = the square root of variance
 - Preferred over variance as it is expressed in the same units as the data
- Low standard deviation
 - Data points tend to be very close to the mean
- High standard deviation
 - Data points are spread out over a large range of values

Use R,
Python, or
Excel to
calculate



Standard deviation example

- You have two sets of numbers:
 - A: 15, 15, 15, 14, 16
B: 2, 7, 14, 22, 30
- Both have the same average (15), but different standard deviation.
 - A: 0.71
 - B: 11.27
- Report either as:
 - A: 15 ± 0.71 and
 - B: 15 ± 11.27
 - or
 - APA style:
 - A: $M=15, SD=0.71$
 - B: $M=15, SD=11.27$

Summary: Descriptive Statistics

- Spend 'quality time' investigating your data
- Describe the central tendency:
 - ▶ Frequencies, percentages
 - ▶ Mode, Median, Mean
- Describe the variability:
 - ▶ Min, Max, Range
 - ▶ Standard Deviation, Variance

